

Section 81 — Master Graph Decomposition Mixed Contest

Contest contract

Six new problems mix condensation degrees, Euler reconstruction, bridge theorems, block-cut paths, and dominator sets. Suggested order: A, D, B, C, E, F.

Problem	Main invariant
A	the only possible universal destination is the unique sink SCC
B	every source and sink SCC needs a new incident direction
C	words are labeled directed edges consumed by Hierholzer
D	Robbins' theorem: a connected undirected graph is strongly orientable iff it has no bridge
E	failed-vertex connectivity is the negation of block-cut path membership
F	a vertex dominates exactly the targets whose dominator set contains it

Recognition and discovery

Quantifier or operation	Compression
reachable from every vertex	unique sink SCC
add edges for strong connectivity	source/sink SCC counts
use every labeled transition once	Euler trail
strongly orient every undirected edge	bridges and Robbins
connectivity after one failure	block-cut path
lies on every directed root path	dominators

After SCC contraction, every source needs new indegree and every sink needs new outdegree, giving $\max(\text{sources}, \text{sinks})$ additions unless there is one SCC. For Robbins orientation, a bridge makes strong orientation impossible; without bridges, downward tree edges and upward back edges let every DFS subtree travel both ways.

Undirected “every path” becomes articulation membership in a block-cut tree. Directed rooted “every path” becomes domination. Tiny oracles should use transitive closure, enumerate orientations, delete failed vertices followed by BFS, and enumerate simple root-to-target paths.

Exercises

Problems A–F form the complete mixed contest.